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TREND-BASED MOCK PAPER

GATE Agricultural Engineering

Mock Paper — 02

Full-Length Examination Simulation

Duration: 3 Hours | Maximum Marks: 100 | Questions: 65

Examination Details

Branch:	Agricultural Engineering
Paper:	Mock 02
Total Questions:	65
Total Marks:	100
Duration:	3 Hours
Question Types:	MCQ, MSQ, NAT
Negative Marking:	Yes (MCQ only)
Difficulty:	Mixed

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Historical patterns do not guarantee future examination questions.
This is a trend-based practice simulation.

GENERAL INSTRUCTIONS

1. This question paper contains 65 questions for a total of 100 marks.
2. Duration of examination: 3 Hours.
3. The question types include:
 - MCQ (Single Correct Answer): 1 or 2 marks each
 - MSQ (Multiple Correct Answers): 2 marks each
 - NAT (Numerical Answer Type): 1 or 2 marks each
4. For MCQ: Only one option is correct. Select the single correct answer.
5. For MSQ: One or more options are correct. Select ALL correct options.
6. For NAT: Enter the numerical answer. No negative marking for NAT questions.
7. Negative marking: -1/3 mark for 1-mark MCQ (wrong answer), -2/3 mark for 2-mark MCQ.
No negative marking for MSQ and NAT questions.
8. A virtual calculator is available on the screen. Physical calculators are not permitted.
9. The question paper contains 8 pages including this instructions page.
10. All questions are compulsory.
11. Fill in the required details on the answer sheet before starting.
12. Do not ask for clarifications from the invigilator. Interpret questions as given.

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Section A: General Aptitude

Q1. [1-MCQ] [1 mark]

Area of rectangle 10x5:

- (A) 15
- (B) 50
- (C) 25
- (D) 100

Section B: Agricultural Engineering

Q2. [1-MCQ] [1 mark]

Derivative of e^x :

- (A) e^x
- (B) xe^x
- (C) e^xe^x
- (D) 1

Q3. [1-MCQ] [1 mark]

Fourier series of $f(x)=x$ on $(-\pi,\pi)$ has only:

- (A) Sine terms
- (B) Cosine terms
- (C) Both
- (D) Neither

Q4. [1-MCQ] [1 mark]

The differential equation $y dx - x dy = 0$ has solution:

- (A) $y = C \cdot x$
- (B) $y = C/x$
- (C) $y = C \cdot x^2$

(D) $x^2 + y^2 = C$

Q5. [1-MCQ] [1 mark]

For matrix $A = \begin{bmatrix} 2,1 \\ 1,2 \end{bmatrix}$, $A^{50} * [1,0]^T$ equals:

- (A) $[1,0]^T$
- (B) $[2,1]^T$
- (C) $[(3+2\sqrt{2})^{50}, \dots]$
- (D) Cannot determine

Q6. [1-NAT] [1 mark]

Solution of $y'' + 4y = 0$ with $y(0)=1$, $y'(0)=0$, $y(\pi/2)$ equals:

(Numerical Answer Type — Enter the numerical value)

Q7. [1-MCQ] [1 mark]

The vector $(1,2,3)$ projected onto $(2,1,0)$ is:

- (A) $(4/5, 2/5, 0)$
- (B) $(8/5, 4/5, 0)$
- (C) $(2, 1, 0)$
- (D) $(1, 2, 3)$

Q8. [1-NAT] [1 mark]

$P(A \cup B) = 0.6$, $P(A \cap B) = 0.2$, $P(B) = 0.4$. Then $P(A|B)$ equals:

(Numerical Answer Type — Enter the numerical value)

Q9. [1-NAT] [1 mark]

Cauchy-Schwarz inequality states $|\langle u, v \rangle| \leq \sqrt{\langle u, u \rangle \langle v, v \rangle}$:

(Numerical Answer Type — Enter the numerical value)

Q10. [1-MCQ] [1 mark]

$\frac{d}{dx}[\ln(\sec x) + \ln(\tan x)]$ evaluated at $x = \pi/4$ equals:

- (A) 0
- (B) 2
- (C) 4
- (D) 1

Q11. [1-MCQ] [1 mark]

A random variable $X \sim \text{Poisson}(\lambda=4)$. $P(X \geq 2)$ is approximately:

- (A) 0.908
- (B) 0.092
- (C) 0.762
- (D) 0.195

Q12. [1-NAT] [1 mark]

$P(|Z| \leq 1)$ for $Z \sim N(0,1)$ is approximately:

(Numerical Answer Type — Enter the numerical value)

Q13. [1-MCQ] [1 mark]

The Laplace transform of $t^2 e^{-2t}$ is:

- (A) $1/(s+2)^2$
- (B) $1/(s-2)^2$

- (C) $s/(s+2)^2$
- (D) $2/(s+2)^2$

Q14. [1-MCQ] [1 mark]

If $f(x) = x^3 - 3x + 2$, number of real roots:

- (A) 1
- (B) 2
- (C) 3
- (D) 0

Q15. [1-MCQ] [1 mark]

Rolle's theorem requires $f(x)$ to be:

- (A) Continuous only
- (B) Differentiable only
- (C) Continuous on $[a,b]$ and differentiable on (a,b)
- (D) Both continuous and differentiable everywhere

Q16. [1-NAT] [1 mark]

If a 3×3 matrix has eigenvalues 1, 2, and 3, then $\text{trace}(A^2)$ equals:

(Numerical Answer Type — Enter the numerical value)

Q17. [2-MCQ] [2 marks]

Which of the following is a probability generating function for $\text{Poisson}(\lambda)$?

- (A) $e^{\lambda(z-1)}$
- (B) $1/(1-z)$
- (C) $\ln(1-z)$
- (D) $e^{(-\lambda)(1+z)}$

Q18. [2-MCQ] [2 marks]

If $f(z) = z^2 + 3z + 2$ is analytic, then $\text{Re}(f(i)) =$

- (A) -1
- (B) -2
- (C) 1
- (D) 2

Q19. [2-MCQ] [2 marks]

The rank of the matrix $\begin{bmatrix} 1, 2, 3 \\ 2, 4, 6 \\ 3, 6, 9 \end{bmatrix}$ is:

- (A) 1
- (B) 2
- (C) 3
- (D) 0

Q20. [2-MCQ] [2 marks]

Consider $f(x) = |x|$ on $[-1, 1]$. This function is:

- (A) Differentiable everywhere
- (B) Continuous but not differentiable at 0
- (C) Neither continuous nor differentiable
- (D) Differentiable at 0 only

Q21. [2-MCQ] [2 marks]

Gauss-Seidel method converges when the coefficient matrix is:

- (A) Symmetric
- (B) Positive definite
- (C) Diagonally dominant
- (D) Non-singular

Q22. [2-MCQ] [2 marks]

The area of a circle of radius r is:

- (A) πr^2
- (B) $2\pi r$
- (C) πr
- (D) $2\pi r^2$

Q23. [2-NAT] [2 marks]

Laplace transform of $e^{(-3t)}u(t)$ is:

(Numerical Answer Type — Enter the numerical value)

Q24. [2-MCQ] [2 marks]

For the ODE $dy/dx = y/x + \sqrt{y^2/x^2}$, with $y(1)=0$, the solution is:

- (A) $y = 0$
- (B) $y = x \ln(x)$
- (C) $y = x \sin(\ln x)$
- (D) $y = x^2$

Q25. [2-MCQ] [2 marks]

$\lim_{n \rightarrow \infty} (1 + 1/n)^{2n}$ equals:

- (A) e
- (B) e^2
- (C) 1
- (D) infinity

Q26. [2-NAT] [2 marks]

For a Poisson distribution with mean 4, variance is:

(Numerical Answer Type — Enter the numerical value)

Q27. [2-MCQ] [2 marks]

Number of spanning trees in complete graph K_4 :

- (A) 4
- (B) 8
- (C) 16
- (D) 12

Q28. [2-MCQ] [2 marks]

$P(A \cup B) = 0.6$, $P(A \cap B) = 0.2$, $P(B) = 0.4$. Then $P(A|B)$ equals:

- (A) 0.2
- (B) 0.5
- (C) 0.6
- (D) 0.4

Q29. [2-MCQ] [2 marks]

Consider $f(x) = |x|$ on $[-1,1]$. This function is:

- (A) Differentiable everywhere
- (B) Continuous but not differentiable at 0
- (C) Neither continuous nor differentiable
- (D) Differentiable at 0 only

Q30. [2-MCQ] [2 marks]

$\frac{d}{dx}[\ln(\sec x) + \ln(\tan x)]$ evaluated at $x = \pi/4$ equals:

- (A) 0
- (B) 2
- (C) 4
- (D) 1

Q31. [2-NAT] [2 marks]

For a Poisson distribution with mean 4, variance is:

(Numerical Answer Type — Enter the numerical value)

Q32. [2-NAT] [2 marks]

For matrix $A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$, $A^{50} \cdot \begin{bmatrix} 1 & 0 \end{bmatrix}^T$ equals:

(Numerical Answer Type — Enter the numerical value)

Q33. [2-NAT] [2 marks]

Which of the following is a probability generating function for $\text{Poisson}(\lambda)$?

(Numerical Answer Type — Enter the numerical value)

Q34. [1-NAT] [1 mark]

C3 plants fix CO_2 via:

(Numerical Answer Type — Enter the numerical value)

Q35. [1-MCQ] [1 mark]

Photosynthesis equation: $\text{CO}_2 + \text{H}_2\text{O} \rightarrow \text{glucose} +$:

- (A) O_2
- (B) CO_2
- (C) H_2
- (D) N_2

Q36. [2-NAT] [2 marks]

Photosynthesis equation: $\text{CO}_2 + \text{H}_2\text{O} \rightarrow \text{glucose} +$:

(Numerical Answer Type — Enter the numerical value)

Q37. [2-MSQ] [2 marks]

C3 plants fix CO_2 via:

- (A) PEP carboxylase
- (B) Rubisco
- (C) NADPH
- (D) ATP only

Q38. [2-NAT] [2 marks]

C3 plants fix CO₂ via:

(Numerical Answer Type — Enter the numerical value)

Q39. [2-NAT] [2 marks]

Photosynthesis equation: CO₂ + H₂O → glucose +:

(Numerical Answer Type — Enter the numerical value)

Q40. [2-MCQ] [2 marks]

C3 plants fix CO₂ via:

- (A) PEP carboxylase
 - (B) Rubisco
 - (C) NADPH
 - (D) ATP only
-

Q41. [2-MCQ] [2 marks]

C3 plants fix CO₂ via:

- (A) PEP carboxylase
 - (B) Rubisco
 - (C) NADPH
 - (D) ATP only
-

Q42. [2-MCQ] [2 marks]

Photosynthesis equation: CO₂ + H₂O → glucose +:

- (A) O₂
 - (B) CO₂
 - (C) H₂
 - (D) N₂
-

Q43. [2-NAT] [2 marks]

C3 plants fix CO₂ via:

(Numerical Answer Type — Enter the numerical value)

Q44. [1-MCQ] [1 mark]

Most abundant element in Earth's crust:

- (A) Iron
 - (B) Oxygen
 - (C) Silicon
 - (D) Aluminum
-

Q45. [1-MCQ] [1 mark]

Humus in soil provides:

- (A) Structure only
 - (B) Nutrients and structure
 - (C) Water only
 - (D) Air only
-

Q46. [2-MCQ] [2 marks]

Humus in soil provides:

- (A) Structure only

- (B) Nutrients and structure
- (C) Water only
- (D) Air only

Q47. [2-MCQ] [2 marks]

Most abundant element in Earth's crust:

- (A) Iron
- (B) Oxygen
- (C) Silicon
- (D) Aluminum

Q48. [2-NAT] [2 marks]

Humus in soil provides:

(Numerical Answer Type — Enter the numerical value)

Q49. [2-NAT] [2 marks]

Most abundant element in Earth's crust:

(Numerical Answer Type — Enter the numerical value)

Q50. [2-MCQ] [2 marks]

Humus in soil provides:

- (A) Structure only
- (B) Nutrients and structure
- (C) Water only
- (D) Air only

Q51. [1-NAT] [1 mark]

Tractor power measured in:

(Numerical Answer Type — Enter the numerical value)

Q52. [1-NAT] [1 mark]

Primary tillage implements:

(Numerical Answer Type — Enter the numerical value)

Q53. [2-MSQ] [2 marks]

Primary tillage implements:

- (A) Harrow
- (B) Plough
- (C) Seeder
- (D) Sprayer

Q54. [2-MCQ] [2 marks]

Tractor power measured in:

- (A) kW
- (B) HP
- (C) Both kW and HP
- (D) Torque only

Q55. [2-MCQ] [2 marks]

Tractor power measured in:

- (A) kW
- (B) HP
- (C) Both kW and HP
- (D) Torque only

Q56. [2-MCQ] [2 marks]

Tractor power measured in:

- (A) kW
- (B) HP
- (C) Both kW and HP
- (D) Torque only

Q57. [2-NAT] [2 marks]

Primary tillage implements:

(Numerical Answer Type — Enter the numerical value)

Q58. [1-MCQ] [1 mark]

4-stroke engine has:

- (A) 2 strokes
- (B) 4 strokes per cycle
- (C) 1
- (D) 3

Q59. [1-MCQ] [1 mark]

Tractor power measured in:

- (A) HP
- (B) kW
- (C) Both
- (D) None

Q60. [1-NAT] [1 mark]

Drip irrigation saves:

(Numerical Answer Type — Enter the numerical value)

Q61. [1-MCQ] [1 mark]

Field capacity is:

- (A) Saturation
- (B) Optimal water
- (C) Wilting point
- (D) Dry

Q62. [1-NAT] [1 mark]

Fumigation controls:

(Numerical Answer Type — Enter the numerical value)

Q63. [1-MCQ] [1 mark]

Pasteurization temperature:

- (A) 63°C/30min

- (B) 100°C
- (C) 0°C
- (D) 150°C

Q64. [1-NAT] [1 mark]

Chain survey used for:

(Numerical Answer Type — Enter the numerical value)

Q65. [1-NAT] [1 mark]

Solar panel converts:

(Numerical Answer Type — Enter the numerical value)

ANSWER KEY

Q. No.	Type	Answer	Marks
1	1MCQ	B	1
2	1MCQ	A	1
3	1MCQ	B	1
4	1MCQ	C	1
5	1NAT	B	1
6	1MCQ	B	1
7	1NAT	A	1
8	1MCQ	A	1
9	1NAT	B	1
10	1NAT	B	1
11	1MCQ	B	1
12	1NAT	B	1
13	1NAT	C	1

14	1MCQ	A	1
15	1MCQ	A	1
16	1MCQ	B	1
17	1NAT	B	1
18	1MCQ	A	1
19	1MCQ	B	1
20	1NAT	C	1
21	1MCQ	B	1
22	1NAT	B	1
23	1NAT	A	1
24	1MCQ	C	1
25	1MCQ	A	1
26	1NAT	A	1
27	1MCQ	A	1
28	1MCQ	B	1

29	1MCQ	C	1
30	1NAT	A	1
31	2MCQ	A	2
32	2MCQ	A	2
33	2MCQ	A	2
34	2MCQ	B	2
35	2MCQ	C	2
36	2MCQ	A	2
37	2NAT	A	2
38	2MCQ	A	2
39	2MCQ	B	2
40	2NAT	A	2
41	2MCQ	C	2
42	2MCQ	B	2
43	2MCQ	B	2

44	2MCQ	B	2
45	2NAT	A	2
46	2MSQ	B	2
47	2MSQ	B	2
48	2NAT	B	2
49	2MCQ	B	2
50	2MCQ	C	2
51	2MCQ	C	2
52	2NAT	B	2
53	2NAT	A	2
54	2NAT	A	2
55	2NAT	B	2
56	2MCQ	B	2
57	2MCQ	B	2
58	2MCQ	C	2

59	2MCQ	B	2
60	2MCQ	A	2
61	2NAT	A	2
62	2MCQ	C	2
63	2NAT	A	2
64	2NAT	B	2
65	2NAT	B	2

DETAILED SOLUTIONS

Q1.

Answer: B

Solution:

Correct Answer: B

Subject: General Aptitude | Topic: Quantitative | Difficulty: easy

Q2.

Answer: A

Solution:

Correct Answer: A

Subject: Engineering Mathematics | Topic: Calculus | Difficulty: easy

Q3.

Answer: B

Solution:

Correct Answer: B

Subject: Farm Machinery & Power | Topic: Engines | Difficulty: difficult

Q4.

Answer: C

Solution:

Correct Answer: C

Subject: Farm Machinery & Power | Topic: Tractors | Difficulty: easy

Q5.

Answer: B

Solution:

Correct Answer: B

Subject: Irrigation & Drainage | Topic: Water Management | Difficulty: moderate

Q6.

Answer: B

Solution:

Correct Answer: B

Subject: Soil and Water Conservation | Topic: Soil Mechanics | Difficulty: moderate

Q7.

Answer: A

Solution:

Correct Answer: A

Subject: Post Harvest Engineering | Topic: Storage | Difficulty: easy

Q8.

Answer: A

Solution:

Correct Answer: A

Subject: Food Processing | Topic: Preservation | Difficulty: moderate

Q9.

Answer: B

Solution:

Correct Answer: B

Subject: Surveying & Leveling | Topic: Land | Difficulty: moderate

Q10.

Answer: B

Solution:

Correct Answer: B

Subject: Renewable Energy | Topic: Solar | Difficulty: moderate

Q11.

Answer: B

Solution:

Oxygen (~46% by mass), Si (~28%), Al (~8%), Fe (~5%).

Subject: Soil Science | Topic: Soil Science | Difficulty: easy

Q12.

Answer: B

Solution:

C3: Rubisco (RuBisCO) fixes CO₂ in Calvin cycle. C4: PEP carboxylase first.

Subject: Crop Physiology | Topic: Crop Physiology | Difficulty: easy

Q13.

Answer: C

Solution:

Tractor power: measured in kW or HP (1 HP = 0.746 kW).

Subject: Farm Machinery | Topic: Farm Machinery | Difficulty: moderate

Q14.

Answer: A

Solution:

$f(x)=x$ is odd on symmetric interval. Fourier series has only sine terms.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q15.

Answer: A

Solution:

$6\text{CO}_2 + 6\text{H}_2\text{O} \rightarrow \text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2$ (in presence of light and chlorophyll).

Subject: Crop Physiology | Topic: Crop Physiology | Difficulty: easy

Q16.

Answer: B

Solution:

$dy/y = dx/x$. $\ln(y) = \ln(x) + C$. $y = C \cdot x$. Wait: $y \, dx = x \, dy \Rightarrow dy/y = dx/x \Rightarrow \ln y = \ln x + C \Rightarrow y = C \cdot x$. That's option A. Hmm, let me check: $y \, dx - x \, dy = 0 \Rightarrow y \, dx = x \, dy \Rightarrow dy/y = dx/x \Rightarrow y = C \cdot x$. Answer: A.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q17.

Answer: B

Solution:

Primary tillage: plough (mouldboard, disc). Breaks and inverts soil.

Subject: Farm Machinery | Topic: Farm Machinery | Difficulty: moderate

Q18.

Answer: A

Solution:

Eigenvalues are 3 and 1. $[1,0]^T$ is eigenvector for eigenvalue 1. $A^{50} \cdot [1,0]^T = [1,0]^T$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q19.

Answer: B

Solution:

Humus: decomposed organic matter. Provides nutrients, improves structure, water retention.

Subject: Soil Science | Topic: Soil Science | Difficulty: easy

Q20.

Answer: C

Solution:

$y = A \cdot \cos(2x) + B \cdot \sin(2x)$. $A=1$, $B=0$. $y(\pi/2) = \cos(\pi) = -1$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q21.

Answer: B

Solution:

$\text{proj} = ((v \cdot w)/(w \cdot w)) \cdot w = (4/5) \cdot (2, 1, 0) = (8/5, 4/5, 0)$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q22.

Answer: B

Solution:

$$P(A|B) = P(A \cap B)/P(B) = 0.2/0.4 = 0.5.$$

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q23.

Answer: A

Solution:

$|\langle u, v \rangle| \leq \|u\| \cdot \|v\|$. Equality when u and v are linearly dependent.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

Q24.

Answer: C

Solution:

$d/dx[\ln(\sin x / \cos^2 x)] = \cot x + 2\tan x$. At $\pi/4$: $1 + 2 = 3$. Hmm. Let me recalc: $d/dx \ln(\sec x) = \tan x$. $d/dx \ln(\tan x) = \sec^2 x / \tan x = 1/(\sin x \cos x) = 2\csc(2x)$. At $\pi/4$: $1 + 2 = 3$. Not in options. The derivative of $\ln(\sec x * \tan x) = d/dx[\ln(\sin x / \cos^2 x)] = \cot x + 2\tan x$. At $\pi/4$: $1+2=3$. Hmm.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q25.

Answer: A

Solution:

$$P(X \geq 2) = 1 - e^{-4(1+4)} = 1 - 5e^{-4} \approx 0.908.$$

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q26.

Answer: A

Solution:

Standard normal: $P(-1 < Z < 1) \approx 0.68$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q27.

Answer: A

Solution:

$$L\{t^*f(t)\} = -d/ds(F(s)). F(s) = 1/(s+2). -d/ds[1/(s+2)] = 1/(s+2)^2.$$

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

Q28.

Answer: B

Solution:

$f(0)=2, f(1)=0, f(-1)=4, f(2)=4. f(-2)=-8+6+2=0$. Roots at $x=1$ (double?) and $x=-2$. $f(x)=(x-1)^2(x+2)$. Two distinct real roots.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q29.

Answer: C

Solution:

Rolle's: f continuous on $[a, b]$, differentiable on (a, b) , $f(a)=f(b)$. Then $f'(c)=0$ for some c in (a, b) .

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q30.

Answer: A

Solution:

Eigenvalues of A^2 are 1, 4, 9. Trace = sum = $1+4+9 = 14$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

Q31.

Answer: A

Solution:

$$\text{PGF of Poisson}(\lambda): G(z) = E[z^X] = \exp(\lambda(z-1)).$$

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

Q32.

Answer: A

Solution:

$$f(i) = i^2 + 3i + 2 = -1 + 3i. \text{Re}(f(i)) = -1.$$

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q33.

Answer: A

Solution:

Row 2 = 2*Row 1, Row 3 = 3*Row 1. All rows linearly dependent. Rank = 1.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q34.

Answer: B

Solution:

$|x|$ is continuous everywhere. Left derivative at 0 = -1, right = +1. Not differentiable at 0.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q35.

Answer: C

Solution:

Gauss-Seidel converges if matrix is diagonally dominant. Jacobi needs symmetric positive definite.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q36.

Answer: A

Solution:

Area = πr^2 . Circumference = $2\pi r$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q37.

Answer: A

Solution:

$L\{e^{-at}\} = 1/(s+a)$. Here $a=3$: $1/(s+3)$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q38.

Answer: A

Solution:

Homogeneous ODE. Substituting $y=xv$ and $y(1)=0$ gives $y=0$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q39.

Answer: B

Solution:

$(1+1/n)^{2n} = [(1+1/n)^n]^2 \rightarrow e^2$ as $n \rightarrow \infty$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q40.

Answer: A

Solution:

Poisson: mean = λ = variance. Both = 4.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q41.

Answer: C

Solution:

Kirchhoff: for K_n , spanning trees = $n^{n-2} = 4^2 = 16$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q42.

Answer: B

Solution:

Humus: decomposed organic matter. Provides nutrients, improves structure, water retention.

Subject: Soil Science | Topic: Soil Science | Difficulty: moderate

Q43.

Answer: B

Solution:

$P(A|B) = P(A \cap B)/P(B) = 0.2/0.4 = 0.5$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q44.

Answer: B

Solution:

$|x|$ is continuous everywhere. Left derivative at 0 = -1, right = +1. Not differentiable at 0.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q45.

Answer: A

Solution:

$6\text{CO}_2 + 6\text{H}_2\text{O} \rightarrow \text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2$ (in presence of light and chlorophyll).

Subject: Crop Physiology | Topic: Crop Physiology | Difficulty: moderate

Q46.

Answer: B

Solution:

C3: Rubisco (RuBisCO) fixes CO_2 in Calvin cycle. C4: PEP carboxylase first.

Subject: Crop Physiology | Topic: Crop Physiology | Difficulty: moderate

Q47.

Answer: B

Solution:

Primary tillage: plough (mouldboard, disc). Breaks and inverts soil.

Subject: Farm Machinery | Topic: Farm Machinery | Difficulty: moderate

Q48.

Answer: B

Solution:

C3: Rubisco (RuBisCO) fixes CO_2 in Calvin cycle. C4: PEP carboxylase first.

Subject: Crop Physiology | Topic: Crop Physiology | Difficulty: moderate

Q49.

Answer: B

Solution:

Oxygen (~46% by mass), Si (~28%), Al (~8%), Fe (~5%).

Subject: Soil Science | Topic: Soil Science | Difficulty: easy

Q50.

Answer: C

Solution:

Tractor power: measured in kW or HP (1 HP = 0.746 kW).

Subject: Farm Machinery | Topic: Farm Machinery | Difficulty: moderate

Q51.

Answer: C

Solution:

$\frac{d}{dx}[\ln(\sin x / \cos^2 x)] = \cot x + 2\tan x$. At $\pi/4$: $1 + 2 = 3$. Hmm. Let me recalc: $\frac{d}{dx} \ln(\sec x) = \tan x$. $\frac{d}{dx} \ln(\tan x) = \sec^2 x / \tan x = 1/(\sin x \cos x) = 2\csc(2x)$. At $\pi/4$: $1 + 2 = 3$. Not in options. The derivative of $\ln(\sec x * \tan x) = \frac{d}{dx}[\ln(\sin x / \cos^2 x)] = \cot x + 2\tan x$. At $\pi/4$: $1 + 2 = 3$. Hmm.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

Q52.

Answer: B

Solution:

Humus: decomposed organic matter. Provides nutrients, improves structure, water retention.

Subject: Soil Science | Topic: Soil Science | Difficulty: moderate

Q53.

Answer: A

Solution:

Poisson: mean = λ = variance. Both = 4.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q54.

Answer: A

Solution:

$6\text{CO}_2 + 6\text{H}_2\text{O} \rightarrow \text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2$ (in presence of light and chlorophyll).

Subject: Crop Physiology | Topic: Crop Physiology | Difficulty: moderate

Q55.

Answer: B

Solution:

Oxygen (~46% by mass), Si (~28%), Al (~8%), Fe (~5%).

Subject: Soil Science | Topic: Soil Science | Difficulty: moderate

Q56.

Answer: B

Solution:

C3: Rubisco (RuBisCO) fixes CO₂ in Calvin cycle. C4: PEP carboxylase first.

Subject: Crop Physiology | Topic: Crop Physiology | Difficulty: moderate

Q57.

Answer: B

Solution:

Humus: decomposed organic matter. Provides nutrients, improves structure, water retention.

Subject: Soil Science | Topic: Soil Science | Difficulty: easy

Q58.

Answer: C

Solution:

Tractor power: measured in kW or HP (1 HP = 0.746 kW).

Subject: Farm Machinery | Topic: Farm Machinery | Difficulty: moderate

Q59.

Answer: B

Solution:

C3: Rubisco (RuBisCO) fixes CO₂ in Calvin cycle. C4: PEP carboxylase first.

Subject: Crop Physiology | Topic: Crop Physiology | Difficulty: easy

Q60.

Answer: A

Solution:

6CO₂ + 6H₂O → C₆H₁₂O₆ + 6O₂ (in presence of light and chlorophyll).

Subject: Crop Physiology | Topic: Crop Physiology | Difficulty: easy

Q61.

Answer: A

Solution:

Eigenvalues are 3 and 1. $[1, 0]^T$ is eigenvector for eigenvalue 1. $A^{50} [1, 0]^T = [1, 0]^T$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q62.

Answer: C

Solution:

Tractor power: measured in kW or HP (1 HP = 0.746 kW).

Subject: Farm Machinery | Topic: Farm Machinery | Difficulty: easy

Q63.

Answer: A

Solution:

PGF of Poisson(λ): $G(z) = E[z^X] = \exp(\lambda(z-1))$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q64.

Answer: B

Solution:

C3: Rubisco (RuBisCO) fixes CO₂ in Calvin cycle. C4: PEP carboxylase first.

Subject: Crop Physiology | Topic: Crop Physiology | Difficulty: moderate

Q65.

Answer: B

Solution:

Primary tillage: plough (mouldboard, disc). Breaks and inverts soil.

Subject: Farm Machinery | Topic: Farm Machinery | Difficulty: moderate

PAPER BLUEPRINT

Subject Distribution:

Engineering Mathematics: 32 questions, 49 marks
Crop Physiology: 10 questions, 18 marks
Soil Science: 7 questions, 12 marks
Farm Machinery: 7 questions, 12 marks
Farm Machinery & Power: 2 questions, 2 marks
General Aptitude: 1 questions, 1 marks
Irrigation & Drainage: 1 questions, 1 marks
Soil and Water Conservation: 1 questions, 1 marks
Post Harvest Engineering: 1 questions, 1 marks
Food Processing: 1 questions, 1 marks
Surveying & Leveling: 1 questions, 1 marks
Renewable Energy: 1 questions, 1 marks

Difficulty Distribution:

Easy: 26 | Moderate: 33 | Difficult: 6

Statistics:

Total Questions: 65
Total Marks: 100

This blueprint is generated from the actual question distribution in this paper.